Coaxolotl filter bank - simulation report

Mast-head LNA + 8-way switched filter bank, bias-T powered over coax. branch feat/filter-sim, 2026-10-28.

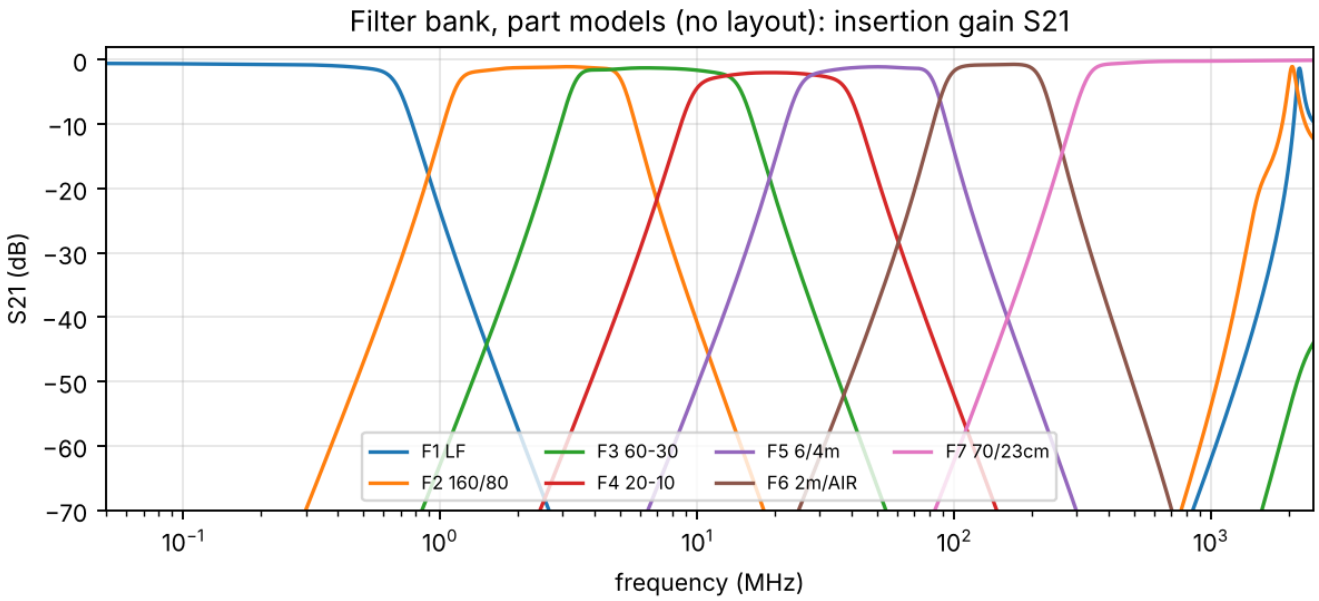
Summary

All eight filter branches were simulated from the KiCad netlist with real part models (vendor S-parameters where published, otherwise datasheet Q, self-resonance and DC resistance). The four VHF/UHF lanes were then simulated as copper with openEMS (FDTD) and the part models connected into those N-ports. **F5, F6 and THRU are fine with the layout. F7 (70/23cm high-pass) is not:** the pad capacitance of 0603 parts over the In1 ground plane (0.4-0.9 pF per lane node) pulls its return loss down to 5.4 dB above 1 GHz. A value change alone (no PCB change) fixes most of it: C701 = C703 = 12 pF, C702 = 6.8 pF, L701 = L702 = 10 nH.

Method

Circuit level (scikit-rf, simu/bank_sim.py): every branch is built from the netlist between the two PE42582 switch ports (switches not yet in the chain). Inductors: Murata LQW18 s2p where available, else datasheet model at minimum Q / minimum SRF (pessimistic). C0G capacitors: ESL + loss tangent. **Layout level** (openEMS, z2 twin kicad_sim 2026-10-21, simu/openems/lane_nport.py): the routed board (JLC04161H-7628 stack-up) cropped to one lane, a lumped port on every part pad referenced to the plane under it, copper as PEC sheet, no solder mask, 10 MHz to 1.5/2.5 GHz. The parts are joined across the port pairs afterwards, so values can be changed without a new field run. GPUs: GTX 1080 (office-z170) and GTX 1080 Ti (X99); one N-port can be split over both machines (--exc / --collect, verified bit-identical).

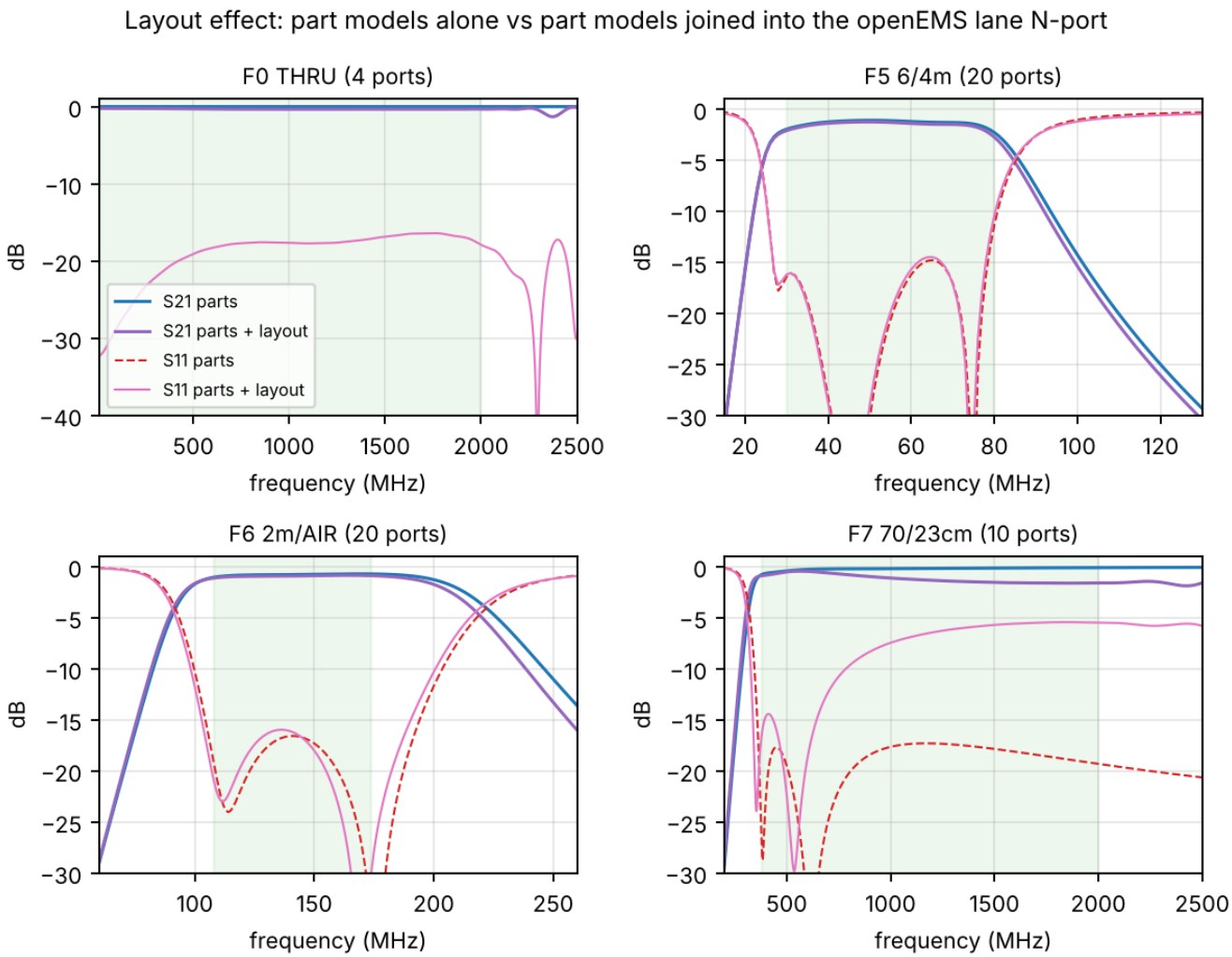
1. Filter bank with part models



branch	name	filter	parts	in-band IL (dB)	min RL (dB)
F1	LF	LPF 550 kHz	5	0.63 - 1.45	17.5
F2	160/80	BPF 1.6-4.0 MHz	10	1.08 - 1.52	13.1
F3	60-30	BPF 4-12 MHz	10	1.27 - 2.03	12.3
F4	20-10	BPF 12-30 MHz	10	2.00 - 2.71	14.1
F5	6/4m	BPF 30-80 MHz	10	1.10 - 2.17	12.3
F6	2m/AIR	BPF 108-174 MHz	10	0.72 - 1.03	16.6
F7	70/23cm	HPF 380 MHz	5	0.10 - 0.78	17.3

IL: insertion loss, RL: return loss, worst case over the branch's band. F4: LQW18CA R33 (L403/L404) is a ferrite-core audio-range choke; LQW18AN R33 recommended (it also has no JLC/EasyEDA footprint). F5 RL: E12 rounding of the design values. F1/F2: stop band returns near 1-2 GHz from inductor self-resonance.

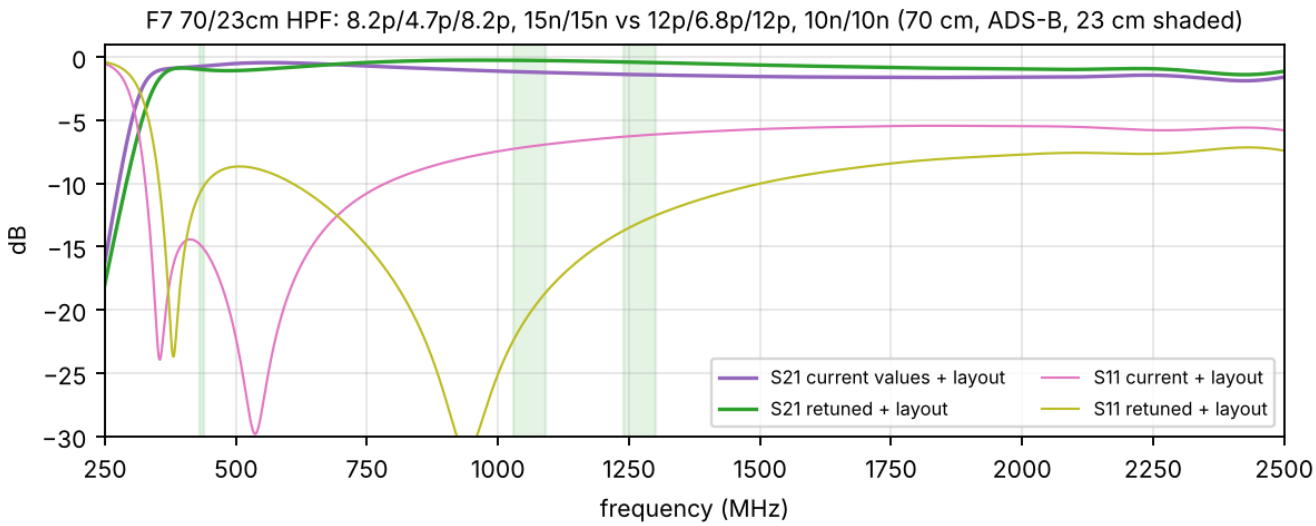
2. Layout effect (openEMS)



lane	ports	IL parts (dB)	IL + layout (dB)	RL parts (dB)	RL + layout (dB)
F0 THRU	4	0.00 - 0.00	0.23 - 0.34	-	16.4
F5 6/4m	20	1.10 - 2.30	1.25 - 2.75	11.3	10.4
F6 2m/AIR	20	0.72 - 1.05	0.85 - 1.12	16.6	16.0
F7 70/23cm	10	0.10 - 0.78	0.45 - 1.63	17.3	5.4

F6 ran on the routed board before the 2 mm lamp-strip change (same lane geometry); F5, F7 and THRU on the final board 6cacac4. THRU: 17 mm of 50 ohm CPWG + R21 between the switch pins (R21's model gives the small loss). Lane nodes carry 1.0-1.1 pF (F5), 0.8-0.95 pF (F6) and 0.4-0.9 pF (F7) to In1; harmless below 200 MHz.

3. F7 70/23cm: retune for the pad capacitance



band	IL current (dB)	RL current (dB)	IL retuned (dB)	RL retuned (dB)
70 cm	0.75	14.8	1.02	10.1
ADS-B 1090 MHz	1.23	6.9	0.29	18.7
23 cm	1.43	6.1	0.45	12.6
1.7-2 GHz	1.63	5.4	0.96	7.7
FM 88-108 MHz (stop)	-57.3 dB		-55.4 dB	

E24 capacitor / E12 inductor grid search with the layout N-port: best worst-case RL over 380-2000 MHz with at least 20 dB rejection below 230 MHz. 70 cm loses a little (RL 14.8 to 10.1 dB), ADS-B and 23 cm gain a lot. Clearing In1/In2 under the F7 pads was also simulated (RL 14 dB) but is not conclusive: the ports then reference B.Cu 1.5 mm below and their own ~1 nH enters the series parts; pad-to-pad ports would be needed.

4. Limits and next steps

- Copper is lossless (PEC) and the solder mask is left out: real IL is slightly higher.
- PE42582 switch S-parameters are not yet in the chain (adds its own IL at both ends of every branch).
- F1-F4 (below 30 MHz) are circuit-level only: the layout is electrically tiny there.
- Decision needed: F7 retune values; F4 L403/L404 replacement; LCSC codes for 14 open BOM lines.